Confidence-Shielded Evaluator-Conditioned Fusion for Risk-Controlled Infrared–Visible Object Detection

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Abstract

Infrared–visible fusion utility is usually assumed to benefit any downstream detector. In practice, the same fused image may improve one detector and degrade another because utility depends on detector representation and confidence calibration. We introduce Confidence-Shielded Evaluator-Conditioned Fusion (CS-ECF), a label-free policy that evaluates visible, CDF, PIAFusion, and SPGFusion candidates with the target detector. CS-ECF averages each candidate's five highest detection confidences and uses a fused input only when its score reaches 1.5 times the visible score; otherwise, it falls back to the visible image. The margin was frozen before testing.

We evaluated CS-ECF on a fixed-split LLVIP test set, grouped six-class M3FD, and the official DroneVehicle test subset, comprising 12 detector checkpoints. Nine checkpoints improved over the visible input, with a dataset-balanced AP50 gain of 0.0133 and a training-run-by-group 95% interval of [0.0026, 0.0261]. Against the stronger Always CDF baseline, the balanced difference was 0.0102 [-0.00057, 0.02980], with no interval separation from zero; the point estimate was concentrated in M3FD and DroneVehicle and was negative on LLVIP. Removing the shield increased the worst AP50 loss from 0.0040 to 0.0584. On two unused zero-shot evaluators, gains over visible were 0.021–0.025 AP50, but differences from CDF crossed zero. Correlated DroneVehicle checkpoints improved by 0.0251 and 0.0293. M3FD gains were concentrated in the people class, and LLVIP intervals crossed zero. These results support risk-controlled evaluator-conditioned fusion, not uniform class improvement or dominance over the strongest fixed input. Serial candidate generation and four detector evaluations required approximately 0.71 s per image pair.

Keywords: infrared–visible image fusion, object detection, evaluator-conditioned fusion, confidence calibration, risk control, thermal imaging

1. Introduction

Infrared–visible imaging combines the spatial detail and color of visible sensors with thermal evidence that can remain available when illumination is poor, smoke reduces contrast, or targets have weak visual texture. This complementarity has motivated pixel-, feature-, and network-based fusion methods for surveillance, autonomous perception, and thermal imaging systems. [1–10] In detection-oriented applications, the value of a fused image is not determined by visual appeal alone; it is the change in detector outputs under a specified operating condition.

Current practice often selects one fused representation and evaluates it with one detector on one benchmark. This design assumes that a globally preferred fusion result remains preferable across detector architectures and training seeds. The assumption is fragile because detectors differ in receptive fields, confidence calibration, training data, and sensitivity to high-frequency detail

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[11, 12]. Consequently, a fusion rule that improves one evaluator may reduce the average precision (AP) of another on the same image.

Per-image selection offers a natural response: run several fusion candidates and choose the one with the strongest detector response. Yet unconstrained selection converts small calibration differences into frequent input switches and can produce severe losses when a spurious high-confidence response is mistaken for improved detection. The problem is therefore not only to predict which fused image looks best, but to decide when the evidence is strong enough to depart from the visible-image safety baseline, a decision related to the classical reject option in recognition [13].

Recent infrared–visible fusion and detection studies have optimized shared fusion networks, cross-modal feature fusion, or hierarchical detectors [14–17]. General-purpose detectors provide the evaluator side of this problem [18, 19]. These studies establish the relevance of joint fusion and detection, but they generally train or optimize a detector-fusion system. They do not establish a label-free arbitration rule whose switching threshold is fixed independently of the confirmatory tests and whose behavior is reported at the level of detector checkpoint, image, and scene or frame group.

We formulate infrared–visible fusion selection as evaluator-conditioned input arbitration. CS-ECF evaluates four candidates with the target detector, computes a top-five confidence score, and accepts a fused candidate only when its score reaches 1.5 times that of the visible input. The threshold is selected on validation data and frozen before testing. We evaluate the policy on LLVIP, M3FD, and DroneVehicle with model-, image-, and group-aware uncertainty, compare it with fixed fusion and an unshielded selector, and report the cost and failure boundaries. The contributions are:

1. an evaluator-conditioned formulation of fusion-input selection that does not use test labels;
2. a confidence shield that reduces catastrophic switching while retaining positive dataset-balanced gains;
3. a three-dataset confirmation that separates image-level uncertainty, scene/group uncertainty, and checkpoint variability, including explicit evidence that the M3FD gain is concentrated in the people class.

2. Methods

2.1. Problem formulation

Let x_v and x_i denote a registered visible–infrared pair, and let $\mathcal{D}$ denote the target detector checkpoint. A candidate $c \in \mathcal{C}$ produces an input image x_c, where $\mathcal{C} = \{\text{vis}, \text{cdf}, \text{pia}, \text{spg}\}$. The policy returns one candidate per pair without access to ground-truth boxes:

$$g(x_v, x_i; \mathcal{D}) \in \mathcal{C}. \tag{1}$$

The final detection output is $\mathcal{D}(x_{g(x_v,x_i;\mathcal{D})})$. The detector is therefore reused for candidate evaluation and final inference; it is not an external label oracle.

2.2. Fusion candidates

The visible input is always included as a safety candidate. CDF combines a pixel-mean image and an IFCNN image through sign-consistent high-frequency details:

$$d(x) = \begin{cases} \operatorname{sign}(d_m + d_i)\min(|d_m|, |d_i|), & d_m d_i > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

A thermal-validity mask M is formed from the infrared top-hat response, infrared intensity, visible luminance, connected components, and smoothing. The shared detail is injected into the LAB luminance channel:

$$L' = L + \alpha M d. \tag{3}$$

Table 1: Training, validation and confirmatory test partitions. The DroneVehicle counts are not reconstructed locally; its checkpoints use the public training and validation partitions recorded by the released model recipe. The evaluated images come only from the official test archive.

Dataset	Train	Validation	Test	Evaluation groups
LLVIP	1,350	150	300	7 scene groups
M3FD	1,600	400	200	2 frame groups
DroneVehicle	public train	public validation	300	3 frame groups

The frozen settings are $q = 0.95$, $\alpha = 0.20$, and $\sigma = 5$. PIAFusion and SPGFusion are generated with their official model definitions and pretrained weights under the same image-pair protocol. [4, 20, 21]

2.3. Confidence-shielded arbitration

For candidate c, let $s_{c,1} \geq s_{c,2} \geq \cdots$ be detector confidences. We define

$$q_c = \frac{1}{\min(K, n_c)} \sum_{j=1}^{\min(K,n_c)} s_{c,j}, \qquad K = 5, \tag{4}$$

where n_c is the number of detections. Candidates with no detections receive $q_c = -1$. The highest-scoring candidate is $c^* = \arg\max_c q_c$, with ties resolved in favor of visible. The selected input is

$$c = \begin{cases} c^*, & c^* \neq \text{vis and } q_{c^*} \geq 1.5 q_{\text{vis}}, \\ \text{vis}, & \text{otherwise.} \end{cases} \tag{5}$$

The margin 1.5 was selected on validation data before the confirmatory tests. The selector uses no test labels and makes no class-specific decision.

2.4. Datasets and evaluators

LLVIP uses a fixed 1,350/150/300 train/validation/test split [22]. The split was formed from non-empty official training labels and the official 300-image test set. Train/test, validation/test and train/validation stem intersections are all empty; the three partitions contain 4,479, 419 and 744 pedestrian boxes. The eight LLVIP confirmatory checkpoints are YOLO11n trained on visible images, a modality-agnostic YOLO11n control, YOLO26n seeds 0–2, RT-DETR-l, and YOLOv8n seeds 0 and 1. YOLO11n seed 0 is used for each YOLO11n family; four independently trained runs of each family are retained as a seed sensitivity analysis. All best checkpoints were selected using the 150 validation images, not the scored test images.

M3FD contributes a grouped six-class split of 1,600 training, 400 validation and 200 test images [23]. Frames are partitioned by consecutive 100-frame group using a fixed group seed, so adjacent frames do not cross splits. The two frame groups in the test set are specified in the released manifest. One YOLO26n and one YOLOv8n visible-trained six-class evaluator were selected on the validation groups.

DroneVehicle contributes 300 image pairs from three fixed consecutive 100-frame groups of the official test archive [24]. The groups were selected with a predefined seed of 2037 before evaluation. The YOLO11x-OBB best and last checkpoints come from one 150-epoch training run (batch size 32, 640-pixel input, seed 0). Because these checkpoints share training data, they are treated as one model cluster rather than independent architectures. The source archive has 4,430,343,899 bytes and SHA-256 `94BF47E493E7A57FD5D900439F614F339E7B6C7181E4D781BCFEBAEB963FFD8E` to identify the exact release used for the test groups.

Table 2: Evaluator inventory. "Checkpoints" counts only the confirmatory models used in the policy comparison; additional independent runs are retained for sensitivity analysis.

Dataset	Evaluator	Training input	Seed / checkpoint	Checkpoints
LLVIP	YOLO11n	visible	seed 0	1
LLVIP	YOLO11n	visible + infrared	seed 0	1
LLVIP	YOLO26n	visible	seeds 0, 1, 2	3
LLVIP	RT-DETR-l	visible	seed 0	1
LLVIP	YOLOv8n	visible	seeds 0, 1	2
M3FD	YOLO26n	visible, six classes	seed 0	1
M3FD	YOLOv8n	visible, six classes	seed 0	1
Drone-Vehicle	YOLO11xOBB	VIS, vehicle OBB	best/last, one run	2

Detection evaluation.. Predictions use a 640-pixel inference size and a confidence floor of 0.001. For YOLO models, non-maximum suppression uses an IoU threshold of 0.7. At each class and each IoU threshold from 0.50 to 0.95 in steps of 0.05, predicted boxes are assigned to ground-truth boxes by one-to-one maximum-cardinality matching over valid pairs. The resulting true-positive flags and confidence scores are pooled across images within a checkpoint and integrated over the 101-point interpolated precision-recall curve using Ultralytics 8.4.189 [25]. AP50 is the mean over the classes present in the dataset; AP50–95 further averages the ten IoU thresholds.

2.5. Statistical analysis

Point estimates use pooled AP50 after concatenating detections across test images within a checkpoint. Image bootstrap intervals resample images within a checkpoint. Group bootstrap intervals resample scenes or consecutive frame groups and are the primary clustered intervals. The dataset-balanced estimand assigns equal weight to LLVIP, M3FD, and DroneVehicle and resamples checkpoints and groups. Checkpoints from the same training run are kept in one cluster. M3FD has only two test groups and LLVIP has seven scenes, so the group bootstrap is reported as a finite-cluster analysis. The training-run-by-group bootstrap used 500 resamples for each checkpoint contrast and 5000 aggregate resamples after group-level contrasts were computed, with `numpy.random.default_rng(2219)` in Python 3.10.22 and NumPy 2.2.6. Spearman association tests used SciPy 1.15.3. The Supplementary Material provides the complete split counts, all per-checkpoint CS-ECF and Always CDF results, the validation-policy record, and the thermal controls. The public analysis script reconstructs each checkpoint and rejects it if its AP does not agree with the published aggregate table before computing the paired interval.

2.6. Implementation and cost

At a standardized fusion input of 384×512, FP32 and batch size one on an NVIDIA RTX 2080 Ti, the CDF pipeline required 22.3 ms, PIAFusion 37.7 ms, and SPGFusion 516.3 ms. Four serial YOLO11x-OBB evaluations at an input size of 640 required 130.0 ms, giving an estimated serial total of 706.3 ms. This estimate excludes model loading or swapping, data loading, and final detection post-processing. SPGFusion's additional peak activation memory was 1453.5 MiB; per-stage memory values must not be summed as an end-to-end peak.

3. Results

3.1. Fusion utility is evaluator-dependent

Across the 12 confirmatory checkpoints, the same candidate library produced both positive and negative gains (Fig. 1). CS-ECF improved nine checkpoints. Its dataset-balanced AP50 gain was

0.0133, with a training-run-by-group hierarchical 95% interval of [0.0026, 0.0261]. The unshielded selector improved seven of 12 checkpoints, had a larger dataset-balanced mean of 0.0332, and produced a worst loss of 0.0584.

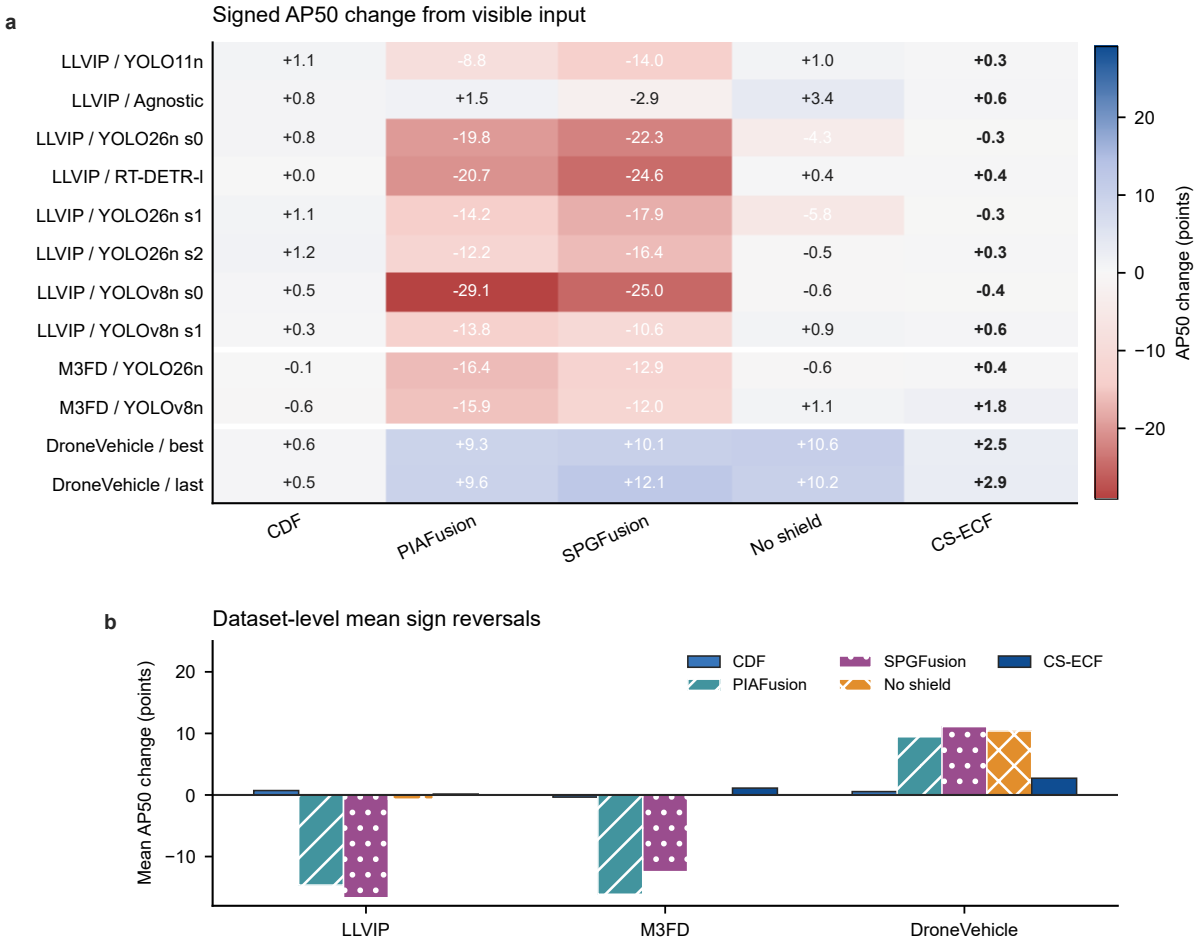


Figure 1: **Evaluator-dependent utility of infrared–visible fusion candidates.** **a**, Signed AP50 change from the visible input for CDF, PIAFusion, SPGFusion, the unshielded selector, and CS-ECF across 12 detector checkpoints. Cells are point contrasts. **b**, Dataset-level mean AP50 changes. Equal candidate sets produce opposite signs across datasets. Source data are provided in `figure1_source.csv`.

3.2. Dataset-specific behavior

On LLVIP, the mean gain was 0.0016 and five of eight checkpoints improved; the model and image intervals crossed zero. On M3FD, both six-class checkpoints improved, with a mean gain of 0.0111. On the independent DroneVehicle test subset, the best and last OBB checkpoints improved by 0.0251 and 0.0293; the combined group interval was [0.0002, 0.0406]. All three DroneVehicle groups were non-negative, but one gained only 0.0002. The two OBB checkpoints originated from the same training run and are not independent architectures.

3.3. Direct comparison with the strongest fixed baseline

Always CDF is the strongest fixed policy, so it is the relevant test of whether evaluator-conditioned arbitration adds value. Across the 12 checkpoints, CS-ECF exceeded Always CDF in six and fell below it in six. The dataset-balanced difference was 0.0102 AP50 in favour of CS-ECF. Its run-cluster by group interval was $[-0.00057, 0.02980]$ and included zero: LLVIP favoured Always CDF, DroneVehicle was represented by a single training-run cluster, and only M3FD had a clearly positive point contrast. Table 4 therefore reports the dataset-specific contrasts rather than only the pooled endpoint.

Table 3: Confirmatory checkpoint summary and policy comparison.

Policy	Positive	Mean	Dataset-balanced	Worst
Always CDF	10/12	+0.0052	+0.0031	-0.0059
Always PIAFusion	3/12	-0.1087	-0.0710	-0.2910
Always SPGFusion	2/12	-0.1136	-0.0602	-0.2501
Top-5 no shield	7/12	+0.0132	+0.0332	-0.0584
CS-ECF	9/12	+0.0074	+0.0133	-0.0040

Table 4: Paired mAP50 difference between CS-ECF and Always CDF. Intervals resample scene or frame groups; the combined interval also resamples training runs within each dataset and then weights datasets equally.

Scope	CS-ECF − Always CDF	95% group-aware interval
LLVIP	−0.0057	[−0.0189, +0.0156]
M3FD	+0.0147	[+0.0100, +0.0159]
DroneVehicle	+0.0217	[−0.0007, +0.0738]
Dataset-balanced	+0.0102	[−0.00057, +0.02980]

The direct comparison changes the claim boundary. CS-ECF is not a statistically separated improvement over Always CDF; its average difference is concentrated in the mixed-class and vehicle-focused evaluations and is negative on LLVIP. The defensible contribution is risk-controlled evaluator-conditioned arbitration and its observed trade-off, not dominance over the strongest fixed input.

3.4. Independent zero-shot evaluator confirmation

To test whether the result was tied to the evaluator families used during policy development, we applied the already frozen selector to two COCO-pretrained models that were not trained, validated, or used to choose the policy: YOLOv9t and YOLOv10s [26, 27]. We retained only their person-class detections and used the same top-five score, 1.5 margin, and four candidate images. Neither model was fine-tuned on LLVIP.

Both unused evaluators improved relative to visible, while neither provided a separated advantage over Always CDF. The external check therefore supports the evaluator-conditioned gain over the visible baseline but not CDF superiority. The shield's tail effect was also mixed: on YOLOv10s it reduced the worst scene loss from 0.0337 to 0.0038, whereas on YOLOv9t the worst scene loss was 0.0167 with the shield and 0.0011 without it. This heterogeneity is reported as a boundary of the risk-control claim.

3.5. The shield is a risk-control mechanism

Without the shield, the policy selected fused candidates on most LLVIP and DroneVehicle images and occasionally produced large confidence gains that did not translate into AP. The 1.5 margin reduced switches and lowered the worst AP50 loss from 0.0584 to 0.0040. On DroneVehicle, however, the shield also reduced the mean gain from 0.1040 to 0.0272. This trade-off is the central quantitative meaning of risk control in this study.

Table 5: Frozen-policy confirmation on two zero-shot evaluators. Values in the second column are visible / Always CDF / no-shield / CS-ECF AP50. Intervals resample LLVIP scene groups.

Evaluator	AP50 (V/CDF/NS/CS)	CS-ECF − visible	CS-ECF − CDF
YOLOv9t	.7698 / .7888 / .8216 / .7948	$+.0250$ $[+.0053, +.0523]$	$+.0060$ $[-.0277, +.0374]$
YOLOv10s	.8344 / .8617 / .8699 / .8557	$+.0213$ $[+.0037, +.0431]$	$-.0059$ $[-.0254, +.0192]$

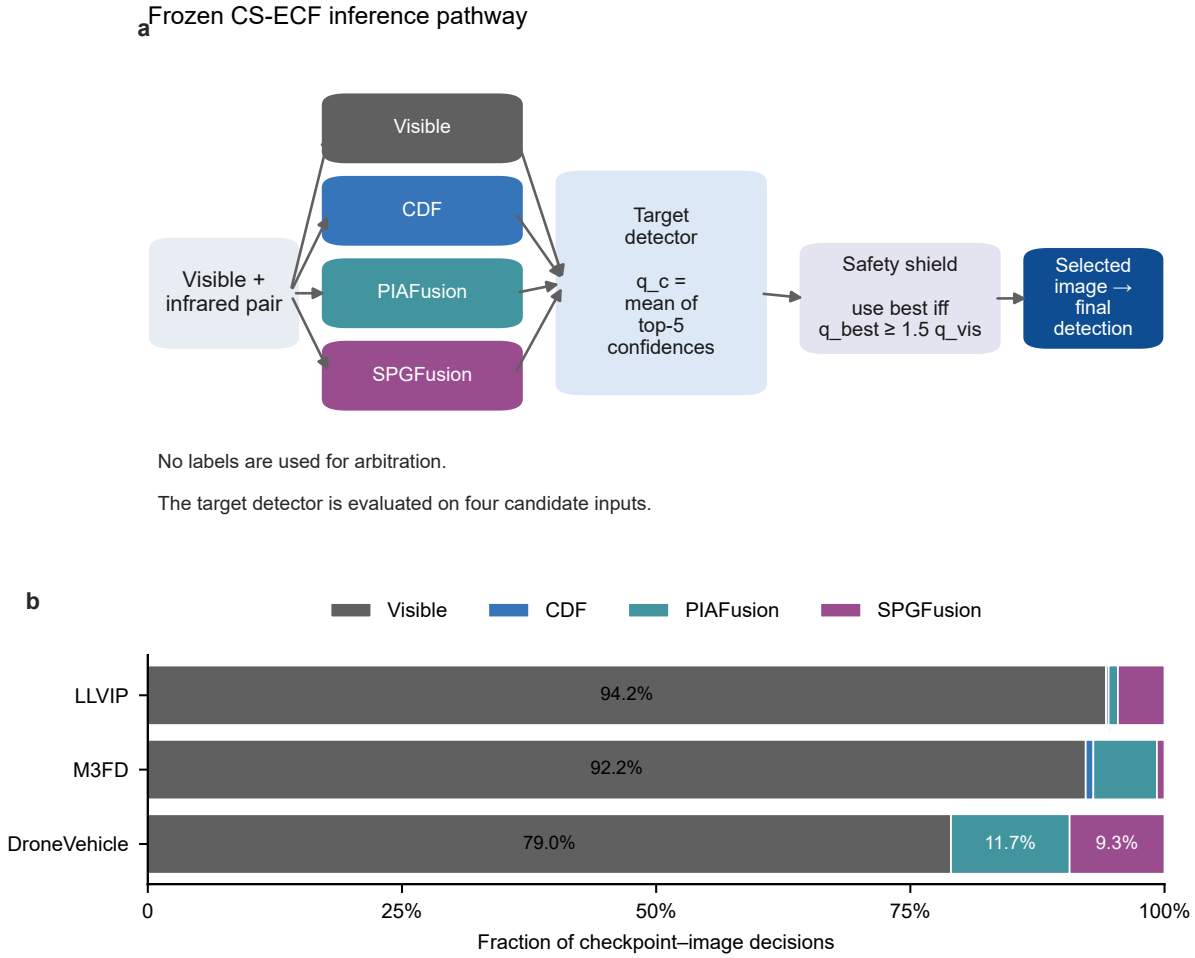


Figure 2: **Confidence-shielded evaluator-conditioned fusion. a**, The visible–infrared pair is passed to visible, CDF, PIAFusion, and SPGFusion candidates. The target detector computes its top-five mean confidence for each candidate. A fused candidate is selected only when its score is at least 1.5 times the visible score; otherwise the visible input is retained. No labels are used for arbitration. **b**, Composition of the frozen decisions across all checkpoint–image trials. Source data are provided in `figure2_source.csv`.

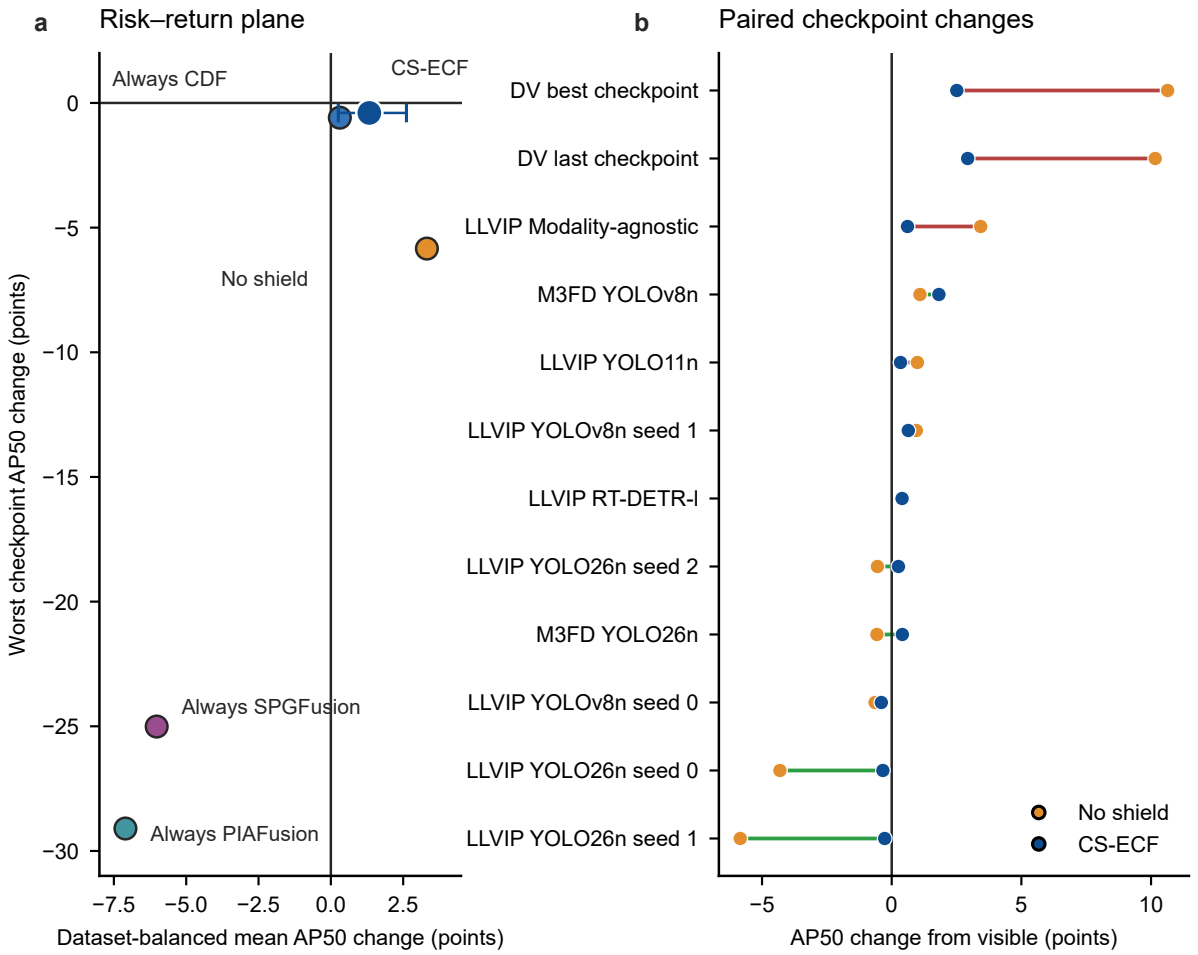


Figure 3: **Risk–return trade-off of fusion-input policies.** **a**, Dataset-balanced mean AP50 change versus worst-checkpoint AP50 change. The CS-ECF horizontal error bar is the training-run-by-group hierarchical 95% interval. **b**, Paired checkpoint changes under the unshielded selector and CS-ECF. The shield sacrifices part of the DroneVehicle mean gain to reduce the worst LLVIP loss. Source data are provided in `figure3_source.csv`.

3.6. Fixed baselines and class concentration

Always CDF improved 10 of 12 checkpoints and had a dataset-balanced gain of 0.0031, but remained negative on both M3FD detectors. Always PIAFusion and Always SPGFusion had negative dataset-balanced gains because their strong DroneVehicle performance did not transfer to the other datasets. In the M3FD six-class evaluation, the CS-ECF gain was concentrated in people: YOLO26n improved by 0.0214 and YOLOv8n by 0.1090, while the other five classes changed by at most 0.0003 in absolute AP50 (Fig. 4). The evidence therefore supports a pedestrian/people-focused interpretation, not uniform six-class improvement.

3.7. Thermal feature association and mask ablation

On LLVIP, infrared top-hat contrast showed a weak positive association with switching (Spearman $\rho = 0.151$, $p = 0.009$), whereas visible luminance was only weakly negative ($\rho = -0.104$, $p = 0.071$). On M3FD, thermal contrast showed a moderate positive association ($\rho = 0.320$, $p < 10^{-5}$). On DroneVehicle, however, both thermal contrast ($\rho = -0.403$) and visible luminance ($\rho = -0.441$) were negatively associated with switching. A high-thermal/low-visible median split did not produce a consistently larger AP gain than the remaining images.

A fixed-input mask ablation gave a smaller but consistent thermal effect on DroneVehicle. Replacing the thermal-validity mask with global detail injection changed the CDF gain from 0.0060 to 0.0057 for the best checkpoint and from 0.0050 to 0.0038 for the last checkpoint. The thermal mask added 0.0007 AP50 on average relative to global injection.

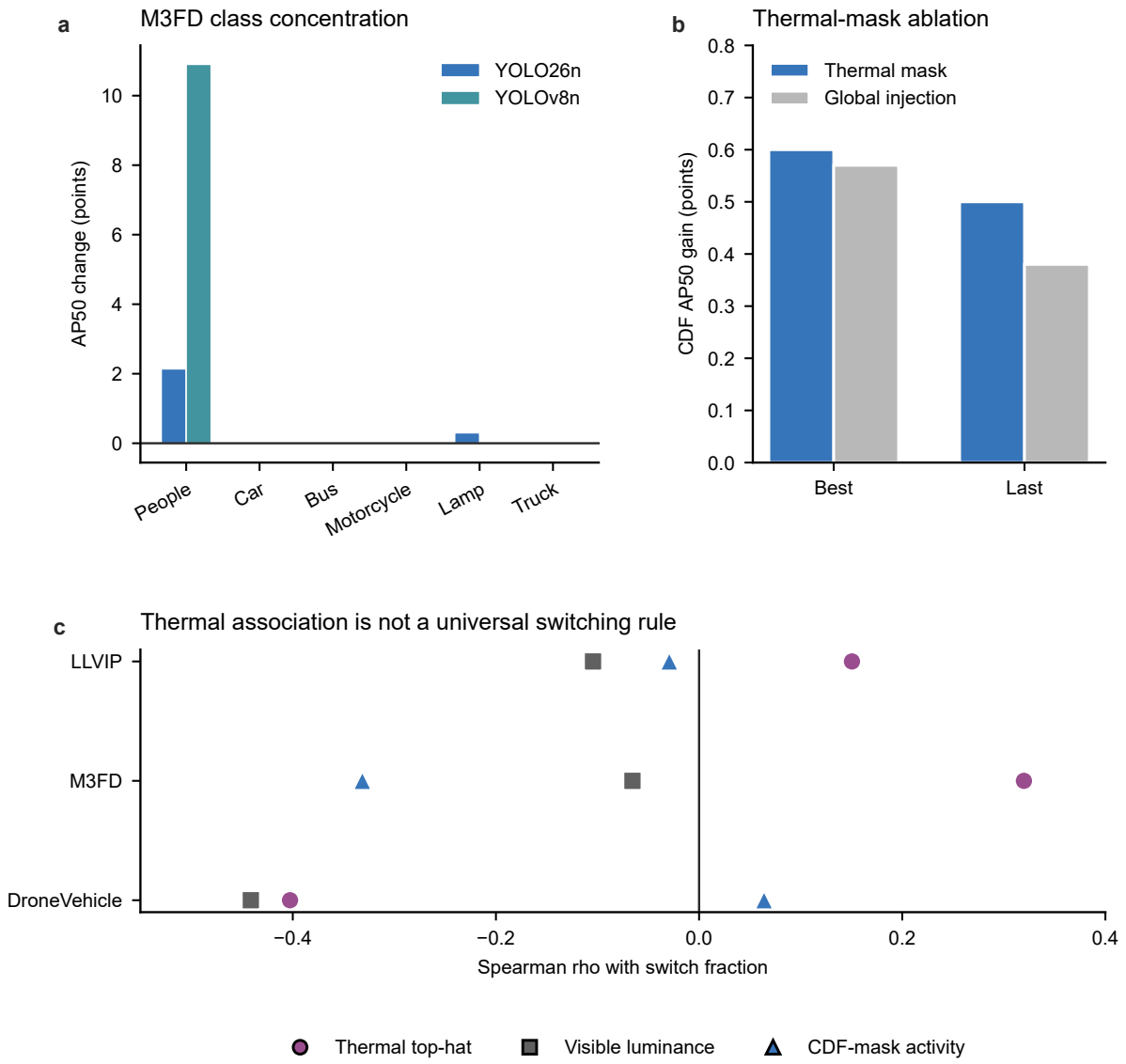


Figure 4: **Thermal constraints and class-specific effects.** **a**, M3FD AP50 changes for six classes. The gain is concentrated in people. **b**, Fixed-input DroneVehicle CDF gain with the thermal-validity mask or global detail injection. **c**, Spearman association between the fused-selection fraction and thermal top-hat contrast, visible luminance, or CDF-mask activity. The association reverses on DroneVehicle, which does not support a universal thermal switching rule. Source data are provided in `figure4_source.csv`.

4. Discussion

The central finding is that fusion utility is not an intrinsic property of a fused image alone. A candidate can improve one detector's confidence ordering while harming another's, and a policy that ignores this dependence can select inputs that appear confident but do not improve AP. CS-ECF responds by using the target detector as an input arbiter and requiring a large visible-relative margin before switching.

The shield's value is clearest in the tail. The unshielded selector achieved a higher dataset-balanced average on DroneVehicle, where both PIAFusion and SPGFusion were strong, but it also produced severe LLVIP losses. The 1.5 margin deliberately favors visible fallback under ambiguous evidence. This is consistent with a deployment rule in which a small probability of a large regression is more costly than a missed opportunity for a moderate gain.

The comparison with Always CDF further bounds the use case. When CDF is known to be stable for a fixed evaluator family, the simpler fixed input remains preferable. CS-ECF is relevant when the deployed detector or data domain can move the best fusion family, as observed between pedestrian, mixed-class and vehicle-focused evaluation. Its small balanced advantage should therefore not be read as a demonstrated replacement for CDF everywhere: the interval included zero.

The two zero-shot checkpoints point in the same direction for the visible baseline: CS-ECF gains of 0.021 and 0.025 AP50 had scene intervals above zero. They again fail to separate CS-ECF from Always CDF. This independent check therefore supports portability of the evaluator-conditioned gain, while reinforcing that the fixed CDF baseline remains competitive.

Two boundaries constrain the interpretation. First, the strongest M3FD effect is class-specific: people detection improves substantially while the remaining classes are nearly unchanged. The selected thermal evidence is therefore most useful for a subset of target appearances, not all six vehicle classes. Second, the DroneVehicle confirmation uses two checkpoints from the same training run. The official test groups are independent of model selection, but the checkpoints share training data and are correlated.

The thermal stratification adds another limitation. Thermal contrast may increase the likelihood of selecting fused evidence in LLVIP and M3FD, but the association reverses on DroneVehicle. Sensor-level thermal evidence and detector confidence therefore capture different information. Neither should be interpreted as a universal utility score. The appropriate claim is thermal fusion utility under a detector-conditioned decision rule, not thermal dominance.

The cost analysis also matters. SPGFusion dominates latency, and four detector evaluations are not free. CS-ECF is best viewed as a risk-controlled system for settings in which the target detector is already run and the cost of a catastrophic input choice is high. It is not a low-latency replacement for a single fusion operator.

5. Conclusions

We showed that infrared–visible fusion utility is evaluator-conditioned and proposed CS-ECF, a confidence-shielded input-arbitration policy that falls back to visible unless a detector-supported fusion candidate clears a fixed visible-relative margin. Across three test datasets and 12 checkpoints, the policy improved nine checkpoints, produced a dataset-balanced AP50 gain of 0.0133, and reduced the worst observed degradation from 0.0584 to 0.0040. The dataset-balanced gain over the stronger Always CDF baseline was 0.0102, with a run-cluster by group interval that included zero. The two unused zero-shot evaluators also improved over visible by 0.021–0.025 AP50, without separating from Always CDF. The evidence supports risk-controlled, pedestrian-centric infrared–visible detection; it does not support uniform six-class improvement, and the additional cost of SPGFusion and four detector evaluations must be considered in deployment.

CRediT authorship contribution statement

Jianyong Zhao: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Zigao Wu:** Supervision, Writing – review & editing. **Mengfei Ma:** Investigation, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Data availability

The underlying datasets are publicly available. LLVIP is distributed at `https://github.com/bupt-ai-cz/LLVIP`. M3FD is distributed with the TarDAL benchmark at `https://github.com/JinyuanLiu-CV/TarDAL`. DroneVehicle is distributed through `https://github.com/VisDrone/DroneVehicle` and was obtained from the public archive `https://huggingface.co/datasets/McCheng/DroneVehicle`. Each dataset is used under the licence stated by its distributor. The fixed DroneVehicle test-group manifest, source-archive size and SHA-256, CS-ECF per-image score files, source-data tables, and scripts that regenerate the reported intervals are included in the public analysis repository described below.

Code availability

The code for reproducing the experiments is available at `https://github.com/lessThanbetter/ssgf` under an MIT licence. The repository contains the fusion and arbitration code, detector-evaluation scripts, fixed split and test manifests, per-image score files, result tables, figure source data, and the numerical checks used to generate the reported intervals. A versioned release will be deposited after acceptance so that the archive and the manuscript describe the same code revision.

Use of generative artificial intelligence

During the preparation of this work the authors used AI-assisted tools to support language editing, data-analysis code development, figure preparation, and the organization of numerical results. The authors reviewed and edited the generated material, verified the reported analyses and figures against the source data, and take full responsibility for the content of the publication.

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